

TIKHAVIN, Russian Federation

WMO: 260940

Lat: 59.650N

Lon: 33.550E

Elev: 200

StdP: 14.59

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.1	-11.2	-22.7	1.6	-17.8	-15.4	2.4	-11.1	14.1	28.8	13.0	29.0	1.4	60	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	83.3	67.8	79.4	65.7	76.0	63.8	69.7	79.9	67.8	76.3	65.9	73.7	5.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.6	73.9	64.6	91.8	71.7	62.8	86.0	70.0	33.8	80.2	32.2	76.3	30.8	74.1	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.7	10.2	-24.3	88.2	8.0	3.8	-30.1	90.9	-34.8	93.1	-39.3	95.2	-45.1	98.0		
(5)			-24.6	72.1	8.0	2.5	-30.3	73.9	-34.9	75.3	-39.4	76.7	-45.2	78.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.0	17.9	17.6	26.7	39.7	50.7	58.8	64.4	60.2	51.2	40.2	29.4	22.1
	DBStd	18.84	13.45	13.57	9.08	8.25	8.44	6.91	6.46	6.15	6.91	7.44	9.88	12.73
	HDD50	4920	996	908	722	325	95	7	1	3	68	313	618	864
	HDD65	9239	1461	1328	1187	758	448	211	89	174	417	769	1068	1329
	CDD50	1279	0	0	0	17	117	271	446	318	102	8	0	0
	CDD65	123	0	0	0	0	5	24	69	24	1	0	0	0
	CDH74	1213	0	0	0	5	70	278	630	219	11	0	0	0
	CDH80	295	0	0	0	0	10	65	175	45	0	0	0	0
	WSAvg	4.8	4.9	4.8	5.2	5.0	5.2	4.7	4.1	4.2	4.4	4.9	5.2	5.3
(15) Precipitation	PrecAvg	30.0	2.4	1.7	1.7	1.6	2.3	3.2	3.2	3.4	2.4	2.6	2.8	2.6
	PrecMax	39.3	3.7	3.0	3.0	2.7	4.3	6.8	9.5	6.9	4.1	4.3	4.2	4.8
	PrecMin	19.7	0.6	0.3	0.3	0.7	0.7	1.0	1.1	0.4	0.7	1.2	0.2	1.2
	PrecStd	4.3	0.8	0.7	0.7	0.6	0.9	1.3	1.8	1.7	0.9	0.9	0.9	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.3	39.9	53.0	72.8	80.8	86.4	90.3	86.0	75.6	61.7	48.5	45.1
		MCWB	37.0	37.4	42.9	56.1	62.2	67.1	70.9	69.4	64.6	55.5	46.3	43.3
	2%	DB	35.9	37.3	45.3	65.9	75.1	80.9	84.5	79.1	69.7	55.5	45.8	40.9
		MCWB	34.8	35.7	38.4	52.1	59.3	66.2	69.0	66.0	61.1	52.2	44.4	39.9
	5%	DB	34.5	35.5	40.8	59.9	70.8	76.7	80.7	75.6	65.1	52.8	43.3	38.4
		MCWB	33.6	34.2	36.0	48.8	56.9	63.2	67.0	64.2	58.5	50.4	42.2	37.4
	10%	DB	33.2	33.8	38.0	54.4	66.6	72.7	76.8	72.1	61.7	50.6	41.0	35.8
		MCWB	32.4	32.8	34.9	45.4	54.4	60.9	65.4	62.5	56.5	48.8	40.0	34.8
	0.4%	WB	37.5	38.4	44.0	57.4	67.2	69.8	72.8	72.4	66.1	56.0	47.0	43.7
		MCDB	38.1	39.1	52.3	71.0	76.9	81.2	85.4	82.6	73.4	60.0	48.0	45.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.2	35.9	39.5	53.3	61.8	67.6	70.6	68.3	62.4	52.8	44.5	40.1
		MCDB	35.7	36.7	43.2	63.9	70.2	79.0	81.8	75.9	67.6	54.7	45.4	40.7
	5%	WB	33.8	34.3	37.1	50.1	58.5	65.2	68.6	65.9	59.5	50.9	42.3	37.5
		MCDB	34.3	35.2	39.6	58.0	67.3	74.2	77.4	73.1	63.7	52.5	43.3	38.3
	10%	WB	32.7	32.9	35.4	46.6	55.9	62.6	66.9	63.7	57.3	48.9	40.1	35.0
		MCDB	33.3	33.8	37.5	53.3	64.7	69.2	74.6	70.0	61.3	50.5	40.9	35.6
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	12.5	14.9	18.4	20.5	18.6	17.9	17.6	14.8	9.4	7.4	8.3
		MCDBR	6.0	7.1	17.7	28.7	27.7	25.2	23.8	24.0	20.2	12.9	7.9	6.3
		MCWBR	6.1	6.4	12.2	16.2	14.1	11.6	10.1	11.9	11.8	9.3	7.4	6.2
	5% WB	MCDBR	5.9	6.9	13.2	25.7	23.5	21.4	21.1	20.0	17.2	11.5	7.6	6.3
		MCWBR	6.1	6.5	9.6	15.0	13.5	10.8	9.8	11.0	10.7	8.9	7.4	6.2
(40) Clear-Sky Solar Irradiance	taub		0.269	0.311	0.331	0.373	0.367	0.362	0.392	0.387	0.355	0.332	0.300	0.253
	taud		2.084	2.087	2.140	2.253	2.398	2.436	2.366	2.369	2.440	2.391	2.200	2.056
	Ebn at Noon		157	211	250	261	273	276	263	255	245	211	147	117
	Edh at Noon		15	26	33	36	33	33	34	32	25	19	14	10
(44) All-Sky Solar Radiation	RadAvg		96	294	746	1182	1600	1702	1623	1279	768	308	92	50
	RadStd		16	69	99	145	149	122	205	127	89	49	12	9

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+1.15	N/A	N/A	N/A	N/A	N/A	-340	-410	N/A

Nomenclature: See separate page